Type MS · Predict how a drug, toxin or disease changes synaptic transmission

Answer 1 of 7 · Question 8 part b.ii · 2 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part b.ii in this mark scheme extract

Number					
8	(a)	(i)	arrow pointing from top to bottom (1)		1
		(ii)	• provides energy/ATP (1) • for synthesis of acetylcholine/neurotransmitter (1) • movement of vesicles/exocytosis (1)		3
	(b)	(i)	• impulses can't pass to intercostal muscles/diaphragm (1) • unable to contract and move ribs/increase volume of thorax/chest (1)		2
		(ii)	• prevents breakdown of neurotransmitter/acetylcholine (1) • which can compete with curare for/binds to, receptor sites (1)		2

Total for this question = 8 marks

Type MS · Predict how a drug, toxin or disease changes synaptic transmission

Answer 1 of 7 · Question 8 part b.ii · 2 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part b.ii in this mark scheme extract

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Type MS · Predict how a drug, toxin or disease changes synaptic transmission

Answer 2 of 7 · Question 6 part d · 3 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part d in this mark scheme extract

6(a)	short term is recent event/long term goes back many years; short term very little processing, long term has more processing;		**(2)**
Question Number	Answer	Additional guidance	Mark
6(b)	cerebral hemispheres/cerebral cortex/frontal lobe/cerebrum;		**(1)**
Question Number	Answer	Additional guidance	Mark
6(c)	• language/speaking/talking problems / repetitive actions/confusion; • loss of voluntary muscle function; • loss of intellectual capacity/function; • deterioration in social behaviour;		**(2 max)**
Question Number	Answer	Additional guidance	Mark
6(d)	• blocks synapse/cleft/build-up of plaques; • prevents diffusion of acetylcholine / transmitter/neurones communicating; • blocks post-synaptic receptors/prevents generation of new impulse; • death of neurones;		**(3 max)**

Question Number	Answer	Additional guidance	Mark
6(e)	• inhibits/prevents breakdown of acetylcholine / transmitter; • increases chance of synaptic transmission / neurones communicating/generating new impulse; • reduces death of neurones;		**(2)**
Question Number	Answer	Additional guidance	Mark
6(f)	formation of β-amyloid is caused by incorrect breakdown of APP; build-up/formation of β-amyloid (in neurones);		**(2)**

Total 12 Marks

Type MS · Predict how a drug, toxin or disease changes synaptic transmission

Answer 2 of 7 · Question 6 part d · 3 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part d in this mark scheme extract

Paper 2: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Predict how a drug, toxin or disease changes synaptic transmission

Answer 3 of 7 · Question 6 part e · 2 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part e in this mark scheme extract

6(a)	short term is recent event/long term goes back many years; short term very little processing, long term has more processing;		**(2)**
Question Number	Answer	Additional guidance	Mark
6(b)	cerebral hemispheres/cerebral cortex/frontal lobe/cerebrum;		**(1)**
Question Number	Answer	Additional guidance	Mark
6(c)	• language/speaking/talking problems / repetitive actions/confusion; • loss of voluntary muscle function; • loss of intellectual capacity/function; • deterioration in social behaviour;		**(2 max)**
Question Number	Answer	Additional guidance	Mark
6(d)	• blocks synapse/cleft/build-up of plaques; • prevents diffusion of acetylcholine / transmitter/neurones communicating; • blocks post-synaptic receptors/prevents generation of new impulse; • death of neurones;		**(3 max)**

Question Number	Answer	Additional guidance	Mark
6(e)	• inhibits/prevents breakdown of acetylcholine / transmitter; • increases chance of synaptic transmission / neurones communicating/generating new impulse; • reduces death of neurones;		**(2)**
Question Number	Answer	Additional guidance	Mark
6(f)	formation of β-amyloid is caused by incorrect breakdown of APP; build-up/formation of β-amyloid (in neurones);		**(2)**

Total 12 Marks

Type MS · Predict how a drug, toxin or disease changes synaptic transmission

Answer 3 of 7 · Question 6 part e · 2 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part e in this mark scheme extract

Paper 2: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Predict how a drug, toxin or disease changes synaptic transmission

Answer 4 of 7 · Question 5 part c.iii · 3 marks

2020 NOV · Paper 01 · extract 1 of 2 · find part c.iii in this mark scheme extract

Number			
5 (a)	any three from • rapid response (to a stimulus) (1) • involuntary/automatic (1) • no brain involvement (1) • protective (1)		3 max
(b)	• A = motor (1) • B = relay/association/connector (1) • C = sensory (1)		3
(c) (i)	• fewer branches/dendrons/dendrites (1) • shorter branches/ dendrons/dendrites (1) • fewer knobs/synapses (1)	R axons	3
(ii)	• transmits impulse (1) • across synapse/(synaptic) cleft (1) • stimulates impulse in next neurone (1)		3
(iii)	• secreted by synapses/synaptic knob (1) • fewer synapses/knobs (1) • less secreted by remaining synapses/knobs (1)		3

Total 15 marks

Type MS · Predict how a drug, toxin or disease changes synaptic transmission

Answer 4 of 7 · Question 5 part c.iii · 3 marks

2020 NOV · Paper 01 · extract 2 of 2 · find part c.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Predict how a drug, toxin or disease changes synaptic transmission

Answer 5 of 7 · Question 7 part b · 3 marks

2020 JAN · Paper 02 · find part b in this mark scheme extract

Number				
7 (a) (i)	<table><tr><th>Part</th><th>Function</th><th>Tick</th></tr><tr><td>medulla</td><td>controls heart rate and breathing</td><td>✓</td></tr><tr><td>pituitary gland</td><td>involved in language</td><td></td></tr><tr><td>hypothalamus</td><td>responsible for osmoregulation</td><td>✓</td></tr><tr><td>cerebellum</td><td>controls voluntary movements</td><td>✓</td></tr></table>	Maximum of two marks if all four boxes are ticked.	3
(ii)	• transmit impulses to effector organs/muscles/glands/controls movement/voluntary actions(1)		1
(b)	• lack of dopamine/no dopamine produced / released/release of dopamine inconsistent(1) • less nerve impulses generated/nerve impulses (to muscles/along motor neurones) inconsistent (1) • tremor/rigidity/slowness of movement/imbalance occur(1)	Allow neurotransmitter for dopamine	3

Total for Question 7 = 7 marks

Type MS · Predict how a drug, toxin or disease changes synaptic transmission

Answer 6 of 7 · Question 5 part a.iii · 2 marks

2023 JUN · Paper 02 (regional R) · find part a.iii in this mark scheme extract

Number			
5 (a) (i)	Any two from: • chemical; • diffused from one neurone to another; • allows neurones to communicate / reference to communication between presynaptic and post synaptic neurones;		2
(ii)	• neurotransmitter/dopamine/ diffuses/travels across synapse; • binds with receptor (on postsynaptic membrane)		2
(iii)	Any two from: • increased movement; • increased heart rate/blood pressure; • more energy; • sleeping difficulties; • anxiety/stress; • hallucinations;		2
(b) (i)	Any two from • tremors/random muscular movement / involuntary movement; • slower movement; • speech/writing changes; • impaired posture/balance; • sleeping problems; • rigidity/stiffness; • anxiety/depression/mental health/memory problems;		2
(ii)	• (145 000 ÷ 100) (x 18) = 26 100; • (26 100 + 145 000) = 171 100;	Allow ecf for mp2 - if mp1 has an incorrect value but this value is added to 145 000 to arrive at the correct answer for the calculation award 1 mark Full marks for correct final answer	2

Total for question 5 = 10 marks

Type MS · Predict how a drug, toxin or disease changes synaptic transmission

Answer 7 of 7 · Question 3 part e · 3 marks

2024 JUN · Paper 02 · find part e in this mark scheme extract

Number			
3 (a)	Smoking cigarettes; Causes (lung) cancer/emphysema/bronchitis/heart disease/eq ; OR lack of exercise; obesity/heart disease/high blood pressure/ type ii diabetes/eq;	Allow drugs or named drug and their effect on health for 2 marks.	(2)
(b)	Any two from: • production of bile reduced/affected; • less fat emulsified/fat not emulsified; • lipase less active/lipase unable to digest as much (fat)/slower lipase activity;	Allow bile not produced Allow description of emulsification	(2)
(c)	blurred/misty vision/blindness/less light passing through the lens / less light reaching the retina;		(1)
(d)	cerebrum/cerebellum/ cerebral cortex/cerebral hemispheres/hypothalamus/medulla (oblongata);	Allow any part of the brain that is affected by alcohol	(1)
(e)	Any three from: • reduces release of neurotransmitters; • reduced diffusion across synapse; • less neurotransmitters bind to receptors on post-synaptic membrane; • slows/ fewer transmission of impulses; • slower reaction /reflexes;		(3)
(f)	Any two from • increased heart rate; • increased breathing rate; • pupil dilation; • increase in blood sugar/glucose levels/ increased breakdown of glycogen to glucose; • increase in blood flow to brain/muscles; • increase in (aerobic) respiration (in muscles);		(2)

Total for Question 3 = 11 marks